Traffic Sign Classification Using Real Images

Abstract

This project classifies real traffic-sign photographs into five categories using GTSRB. It compares pretrained ResNet-18, MobileNetV3-Small and EfficientNet-B0 under the same five-epoch budget. Training uses 2,500 images, validation uses 750 images and testing uses 2,820 images. Related training views are separated by sign track. One training track containing exact copies of test images is excluded. ResNet-18 is selected by validation macro-F1, not test accuracy. Its test accuracy is 99.89% and macro-F1 is 0.9992. An exported ONNX model is checked using actual inference. These results concern a five-class cropped-sign task, not full-scene detection or autonomous driving.

1. Introduction and Objective

Traffic signs give drivers instructions. Recognising a sign category is a useful computer-vision task with a clear input and output. The project was narrowed from a detector-plus-classifier demonstration to image classification so that the work could focus on real data, a fair model comparison and understandable evaluation. YOLO is not used in the new experiments; the old synthetic video is retained only as historical work. The research question is: which of three pretrained classifiers performs best under a small, shared training budget, and what are the costs in model size and inference time?

The objective is to compare recognition quality, model size and inference time using the same real-image split. The five output categories are Stop, No entry, Yield, Speed limit 30 and Speed limit 50. A model receives one annotated sign crop and returns five category scores. The highest score supplies its prediction; it does not locate the sign in a full road scene.

The revised task builds on image preprocessing, transfer learning and architecture comparison. ResNet-18 is the baseline, MobileNetV3-Small provides a smaller alternative, and EfficientNet-B0 adds another compact architecture. Training, validation and test roles remain separate. Reported results are measured on real benchmark photographs, not the previous generated images.

2. Methodology

2.1 Data, annotations and exploratory data analysis (EDA)

GTSRB contains 51,839 real images across 43 classes: 39,209 official training images and 12,630 official test images. EDA checked every annotated image: all decoded successfully, with no image-size mismatches or invalid inclusive ROIs. Class sizes range from 210 to 2,250 training images. Median training width and height are both 43 pixels, and brightness varies.

Eight exact train/test image pairs were found in stop-sign track 00023. All 30 training images in that track were removed before sampling. Training and validation use different physical-sign tracks, with 85 training tracks and 25 validation tracks. Seed 42 controls sampling; all official test examples of the five selected categories are retained.

Category	Training	Validation	Test
stop	500	150	270
no entry	500	150	360
yield	500	150	720
speed limit 30	500	150	720
speed limit 50	500	150	750
Total	2500	750	2820

The separate EDA report in report/gtsrb_eda/ contains the full class-count chart, image-size and brightness distributions, real training-image examples, and duplicate file lists.

2.2 Preprocessing and classification

Annotated rectangles supply the sign crops. Inclusive ROI maxima are converted to exclusive crop maxima. Images are resized to 128 by 128 pixels and normalised using ImageNet statistics. Training adds colour jitter and rotations up to eight degrees; validation and testing use fixed preprocessing. Horizontal flips are excluded because they can reverse numbers or sign meanings.

All three classifiers start with ImageNet-pretrained weights and a new five-class output layer. All parameters are fine-tuned for five epochs using cross-entropy and AdamW: learning rate 0.0003, weight decay 0.0001, batch size 16 and seed 42. The same split and preprocessing are used for each model. The 128-pixel input limits CPU computation; other resolutions were not compared.

2.3 Evaluation metrics and experimental design

Accuracy measures the fraction correct. Precision measures how often a predicted category is correct; recall measures how many actual members are found. F1 combines them: 2PR/(P+R). Macro-F1 gives each class equal weight, making it useful with unequal test counts. Checkpoints and the final architecture are selected by validation macro-F1; lower CPU latency breaks an exact tie. Test predictions do not select the model.

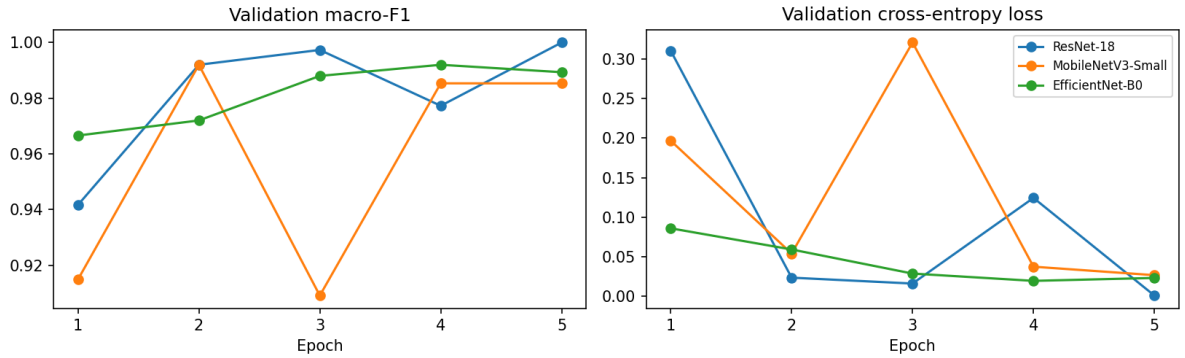
3. Experiments and Results

3.1 Classifier comparison

There are three neural-network experiments, one per architecture and one seed each. A constant training-label baseline predicts no entry after a tie between the balanced training labels; it achieves 12.77% test accuracy and 0.0453 macro-F1 without examining pixels.

Model	Val F1	Test Acc	Macro P	Macro R	Macro F1
ResNet-18	1.0000	99.89%	0.9992	0.9992	0.9992
MobileNetV3-Small	0.9920	95.04%	0.9582	0.9588	0.9581
EfficientNet-B0	0.9920	99.47%	0.9960	0.9945	0.9952

Selected by validation: ResNet-18. Test crop predictions: 2,817 correct and 3 incorrect. The selection is not based on which test accuracy is largest.

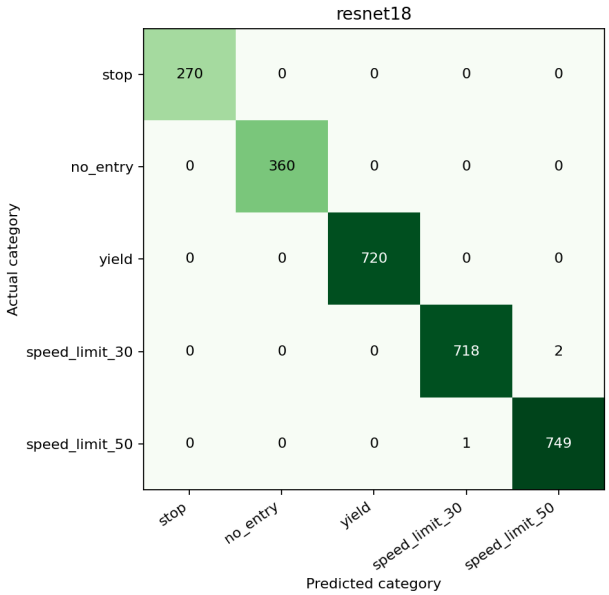


3.2 Class-level results and error analysis

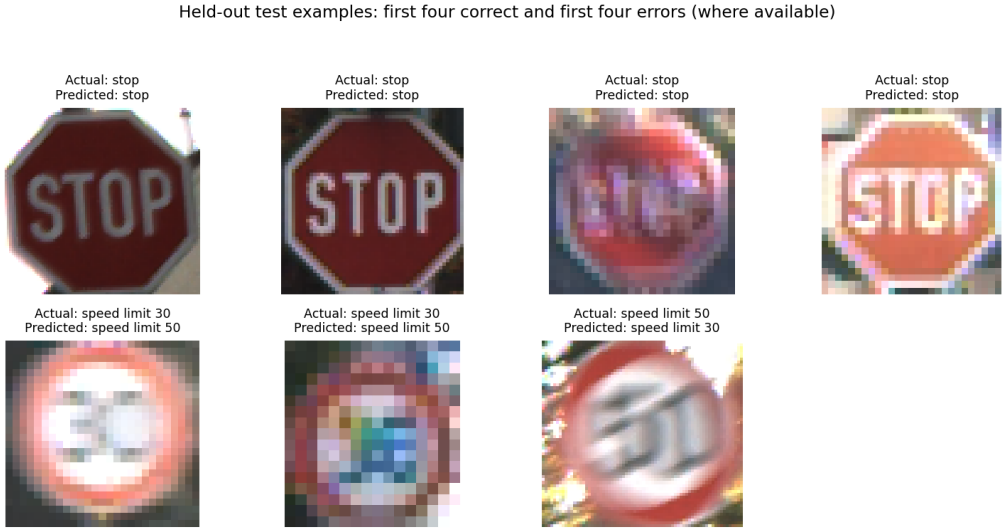
Category	Precision	Recall	F1	Count
stop	1.0000	1.0000	1.0000	270
no entry	1.0000	1.0000	1.0000	360
yield	1.0000	1.0000	1.0000	720
speed limit 30	0.9986	0.9972	0.9979	720
speed limit 50	0.9973	0.9987	0.9980	750

The selected model's three errors are between the two speed-limit categories: two 30 signs are predicted as 50, and one 50 sign is predicted as 30. Stop, No entry and Yield have no errors in this test subset. These are crop-classification results, not evidence of full-road detection accuracy.

3.2 Class-level results and error analysis (continued)



Rows show actual categories; columns show predictions. Diagonal entries are correct. Off-diagonal entries are errors. Counts are not percentages, so larger classes contribute more entries.



The gallery contains the first four correct predictions and the three errors in saved test order. These are held-out real photographs, not a representative random sample. All predictions and image paths are saved in the results CSV. Visual difficulty alone does not prove why a model made an error.

3.3 Timing

Model	Parameters (M)	Forward (ms)	Train (min)
ResNet-18	11.179	6.91	7.23
MobileNetV3-Small	1.523	27.75	8.89
EfficientNet-B0	4.014	79.30	35.36

PyTorch timing uses CPU, two threads, one image, ten warm-ups and 30 repetitions. It excludes reading and preprocessing; training time includes startup and checkpoints. Smaller parameter counts did not guarantee faster CPU inference. This is not a road-video real-time benchmark.

3.4 Export, optimisation and verification

ONNX is an export format, not another trained model. The selected classifier was exported and checked. Actual ONNX Runtime inference was evaluated on all 2,820 test crops. The first 32 images were also compared numerically with PyTorch; maximum absolute logit difference: 0.00000954.

Variant	Size (MB)	Accuracy	Macro-F1	Mean ms
float32	44.71	99.89%	0.9992	9.56
dense_int8	44.71	99.89%	0.9992	8.14

Dynamic INT8 applies only to dense MatMul/Gemm weights; convolution layers stay float32. This is not full-model INT8. ONNX CPU timings use two threads, ten warm-ups and 50 repetitions, excluding preprocessing. Float32 remains the default; optimisation is reported as an experiment, not an automatic improvement.

4. Discussion and Reproducibility

Hydra stores configuration, Lightning organises training, TensorBoard logs curves and DVC records dataset/model versions. Archive URL and SHA-256 provenance and the fixed split hash are saved. Data and weights are local unless a DVC remote is configured. The Dockerfile has not been built here because Docker is not installed. A fork relationship to the recommended Lightning-Hydra template has not been verified.

Practical issues included finding appropriate real categories, matching official CSV headers, identifying train/test duplicates, separating related sign views and managing CPU training time. All 30 images in the duplicated stop-sign training track were removed before sampling. Original synthetic scores are not used as real-data evidence.

These results cover five known German sign categories, with no unknown-category rejection or full-scene detector. One seed and five epochs do not prove a universal architecture ranking. Repeated runs and external real-world images are needed before broader claims.

5. Conclusion

The project compares three pretrained classifiers using real GTSRB photographs, a fixed track-separated development split and held-out testing. ResNet-18 is selected using validation macro-F1 and achieves 99.89% test accuracy and 0.9992 macro-F1. The main result is supported by per-class metrics, saved predictions and verified ONNX inference. The scope remains five-class cropped-sign classification; external testing and repeated runs are needed before making broader claims.

Evidence Sources and References

Measured evidence: report/gtsrb_results/ contains comparison.json, dataset_summary.json, learning_curves.json, selected_test_predictions.csv and onnx_verification.json. Full EDA: report/gtsrb_eda/.

[1] Stallkamp et al., GTSRB, IJCNN 2011. github.com/houbensebastian/GermanTrafficSignBenchmarks. [2] PyTorch torchvision model documentation: docs.pytorch.org/vision/stable/models.html. [3] Scikit-learn classification metrics: scikit-learn.org/stable/modules/model_evaluation.html.